

Social Media Threat Intelligence Engine

Team Name: Team Shin-chan | **Track:** Problem #06 — Track 2: Cyber Forensics (IBM Bob AI Innovation Hackathon / INNOVAI)

Repository: github.com/Tirth-chokshi/bob-ai-hackathon-shin-chan | Submission Date: September 2026

1. Executive Summary & Proposed Solution

During sensitive public incidents (such as communal tensions, civil protests, or natural disasters), malicious actors, hyper-partisan networks, and bot rings deploy Coordinated Inauthentic Behavior (CIB) to rapidly amplify inflammatory rumors, manipulate public opinion, and coordinate offline violence.

Existing cyber cell monitoring workflows rely on manual post-by-post scanning or simple keyword alerts. This approach consistently fails because individual posts often appear harmless in isolation, while working in synchronized concert to spark dangerous real-world unrest.

The Solution

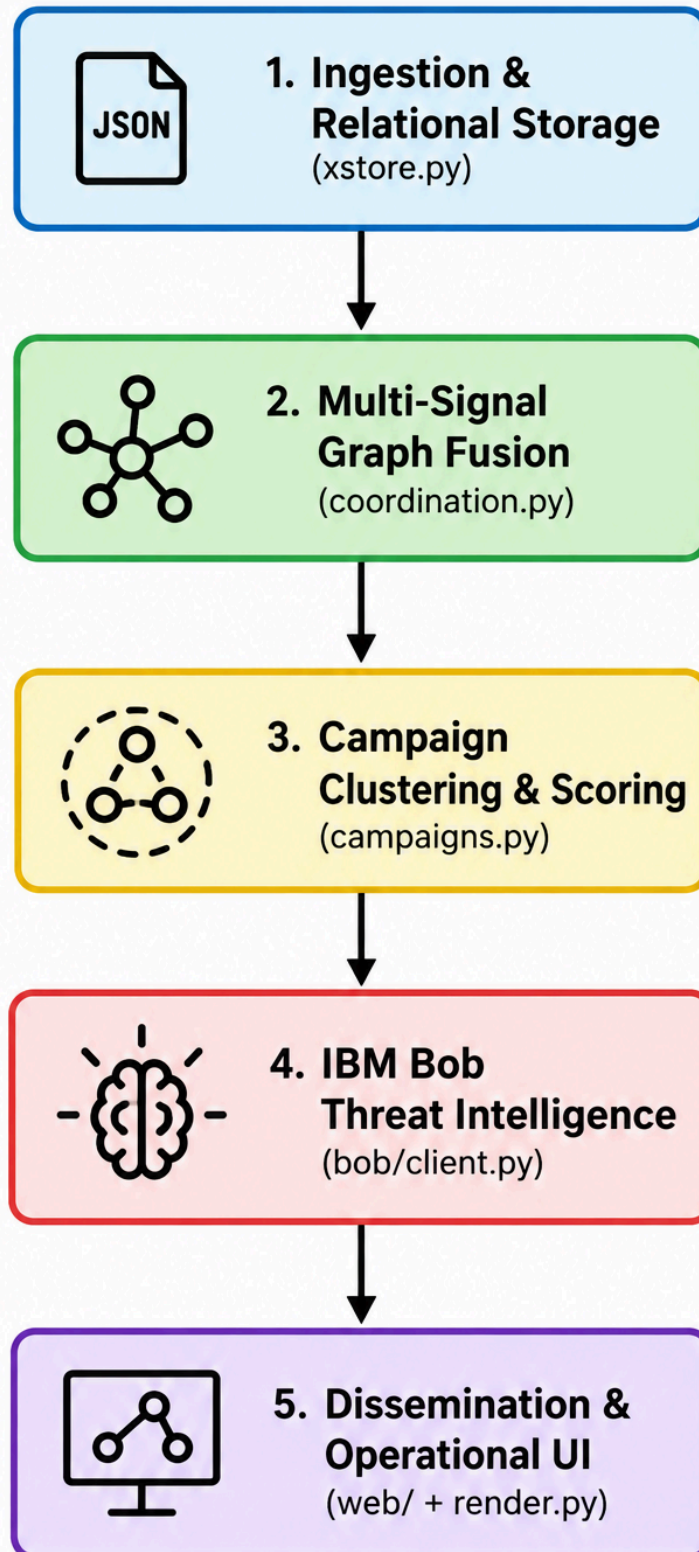
We have built the Social Media Threat Intelligence Engine, an automated, AI-powered cyber-forensics platform that:

- **Prioritizes Behavior Over Content:** Ingests raw social media data (standard X API v2 format) and mathematically identifies accounts acting in locked temporal synchrony within 60-second windows across 5 orthogonal coordination vectors (same text, similar text, shared links, co-retweets, and reply pile-ons).
- **Pinpoints Originators & Amplifiers:** Uncovers the exact seed accounts that ignited viral campaigns (e.g., recycling years-old out-of-state violent footage with false local tags) and traces propagation through high-degree amplifier hubs.
- **Extracts Real-World Physical Threats with IBM Bob:** Uses IBM Bob (Granite LLM) to detect explicit offline gathering calls ("*assemble at Maujpur Chowk at 5:00 PM with sticks*"), estimates threat severity, and automatically maps observed violations to specific Indian legal statutes (**Bharatiya Nyaya Sanhita / BNS 2023** and the **IT Act**).
- **Delivers Actionable Police Dossiers:** Automatically drafts a Station House Officer (SHO) intelligence memorandum complete with SHA-256 evidence hashing for court admissibility under **Section 63 of the Bharatiya Sakshya Adhiniyam (BSA 2023)**.

2. Technical Architecture & Approach

The system follows a modular, 5-stage pipeline designed for speed, reproducibility, and explainability

Key Technical Phases



- **Phase 1: Ingestion & Relational Normalization:** Strict validation of X API v2 responses into an embedded SQLite database (`x.db`), normalizing posts, user profiles, place geolocations, and media attachments into a clean `posts` relational view.
- **Phase 2: Multi-Signal Coordination Detection:** Evaluates pairwise account synchrony within 60-second sliding windows across 5 networks (`co_tweet`, `co_similar_tweet`, `co_link`, `co_retweet`, and `co_reply`). Multi-signal edges are accumulated into an undirected graph, filtering out organic coincidences.
- **Phase 3: Louvain Community Partitioning & CIB Scoring:** Applies deterministic Louvain modularity to group coordinated accounts into discrete campaigns (`$c_1, c_2, \dots$`). Each campaign receives an explainable 0–100 CIB score based on Speed (25%), Text Duplication (25%), Multi-Signal Diversity (15%), Fresh Account Share (15%), Burst Volume (10%), and Target Concentration (10%).
- **Phase 4: IBM Bob Reasoning & Legal Mapping:** A representative spread sample of posts is passed to IBM Bob via headless CLI. Bob classifies threat type, determines severity (1–5), extracts planned offline gathering parameters (`where`, `at`, `what`), and maps offenses to validated statutory tables (BNS 196, BNS 353, IT Act 66D).
- **Phase 5: Visual Investigation & Brief Generation:** Renders an interactive Cytoscape network graph, a 3-stage propagation flow (*Origin Seeds* $\rightarrow$ *Amplification Hubs* $\rightarrow$ *Public Impact*), an authentic Twitter/X forensic feed, and a single-click print-ready intelligence dossier.

3. Key Tools, Technologies & Sources

Component / Layer	Technology / Tool	Version / Source	Purpose & Role
AI Reasoning & LLM	IBM Bob Shell 2.0 (Granite LLM)	IBM Innovation Hackathon (<code>npm/bobshell</code> , headless CLI <code>bob run</code>)	Autonomous qualitative threat classification, offline physical event extraction, and legal section mapping.
Agent Tool Protocol	FastMCP	GitHub / PyPI (<code>fastmcp</code>)	Model Context Protocol server exposing graph queries, post evidence, and campaign data to Bob.

Component / Layer	Technology / Tool	Version / Source	Purpose & Role
Coordination Engine	Coordination Network Toolkit	QUT Digital Observatory (Queensland University of Technology, PyPI / GitHub, MIT License)	Core algorithms for detecting temporal co-occurrences (co-tweet, co-retweet, co-link, co-reply).
Graph Algorithms	NetworkX	PyPI (networkx)	Louvain community modularity optimization, graph construction, and degree centrality calculation.
Backend Framework	FastAPI & Uvicorn	PyPI (fastapi , uvicorn)	High-performance asynchronous Python REST API serving datasets, graphs, and live analysis streams.
Data Validation	Pydantic v2	PyPI (pydantic)	Strict data typing and schema validation for API inputs, outputs, and Bob structured verdicts.
Local Relational Storage	SQLite 3	Python Standard Library	Zero-dependency, embedded relational database (x.db) for storing ingested X API v2 records.
Frontend Framework	React 19 & Vite	npm (react , react-dom , vite)	Fast, reactive Single Page Application (SPA) dashboard architecture.
Network Visualization	Cytoscape.js	npm (cytoscape)	High-performance, GPU-accelerated interactive network graph canvas with dynamic layouts.
Styling & Design System	Tailwind CSS v4 & Lucide Icons	npm (tailwindcss , lucide-react)	Responsive, high-contrast cyber-forensic design system adhering to strict accessibility standards.

4. Key Results & Forensic Impact

- **Actionable Lead Time:** Detects coordinated bot mobilization hours before physical crowds converge, giving field police actionable warning.
- **Explainable AI Guardrails:** 100% transparent CIB score breakdown; every legal section and physical gathering quote extracted by IBM Bob is verified in code against database rows, eliminating LLM hallucinations.
- **Court-Admissible Evidence:** Implements SHA-256 evidence hashing for strict compliance with Section 63 of the **Bharatiya Sakshya Adhiniyam (BSA 2023)**.
- **Operational Efficiency:** Processes thousands of posts into a structured Station House Officer briefing dossier in under 60 seconds at a minimal cost of ~0.03–0.04 Bobcoins.

5. Team Shin-chan

- **Tirth Chokshi** (Lead / Forensics) — System architecture, 5-signal coordination engine, and CIB scoring pipeline.
- **Milind Pawar** (Frontend Engineering) — React 19 UI, Cytoscape network graph, and 3-stage propagation flow.
- **Jainik Devada** (Bob Layer & MCP) — IBM Bob CLI orchestration, BNS legal mapping.
- **Jigar Jariwala** (Data & Benchmarking) — gathering mock datasets, historical riot benchmarks, and evaluation.